

How Vague Words Shape Interpretation: Evidence from Approximators

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Abstract

Human magnitude estimates are typically approximate. Whether they arise from observation with noise or from inferential processes, they tend to give rise to subjective probability distributions that are peaked around a central value (Borel, 2014; Mamassian et al., 2002; Tenenbaum & Griffiths, 2001). In this paper, we test the hypothesis that *vague expressions* are particularly well-suited to communicate such approximate magnitude estimates. We specifically focus on vague approximators (“around”). In previous work (Égré et al., 2023), we developed a Bayesian account of the interpretation of such expressions which predicts that regardless of the prior used, the posterior distribution of possible numbers induced by an expression of the form “around n ” should be more peaked toward its central value n than the posterior induced by a precise counterpart of the form “between m and m' ”. The underlying interpretive mechanism (Bayesian inference about the unknown interval denoted by “around n ”) is partly similar to what has been proposed for generalization inferences from exemplars (Tenenbaum & Griffiths, 2001). In three experiments, we use a novel probability-elicitation paradigm to compare the posterior distributions that are evoked by the vague approximator “around” with those that result from the precise approximator “between”. While both are used to communicate numerical uncertainty, our results confirm that “around” gives rise to a posterior distribution whose shape is more peaked than the one produced by a corresponding “between”-statement.

Keywords: Vagueness ; Approximation ; Generalization; Probability ; Lexical Uncertainty; Bayesianism; Hypothesis Averaging

1 Introduction

The categorization of what we perceive typically involves building an approximate representation of determinate magnitudes. This holds for how we represent continuous magnitudes, such as loudness (Stevens, 1956), but also for discrete magnitudes, such as numerosities (Feigenson et al., 2004), and similarly for countless dimensions of our experience. For most of our interactions, for example, we usually don’t know our interlocutor’s age or height, and we couldn’t precisely quantify the hair mass on their scalp, but we can form relative estimates. In such cases, our estimates are centered on focal values that we take to be more likely than others (Borel, 2014; Mamassian et al., 2002).

How does language allow us to communicate those epistemic states, which are centered on focal values without being fully determinate? The answer to this question concerns *vague words*. Vague words are words like *tall*, *old*, or *bald* (see, a.o., Fine, 1975; Hersh and Caramazza, 1976; Russell, 1923; Smith, 2008; Verheyen and Égré, 2018; Williamson, 1994). They describe exact quantities in a qualitative manner, without communicating any exact measurements, and their use seems inherently tied to the fact that we represent magnitudes in ways that are analog and probabilistic, rather than discrete and categorical (Égré, 2017). Despite that, they allow us to judge some values to be more probable than others.

In recent years, some attention has been given to the interpretation of these vague expressions in the psycholinguistic literature, notably in relation to Bayesian models of communication (Égré et al., 2023; Frank & Goodman, 2012; Frazee & Beaver, 2010; Goodman & Frank, 2016; Lassiter & Goodman, 2013; Qing & Franke, 2014; Sutton, 2018). It has been argued, in particular, that while vague adjectives like “tall”, “old”, and “bald” do not draw determinate boundaries along the relevant scales, they nevertheless produce informative updates on the listener, by modifying their prior probabilities in directions that depend on the lexical meaning of those words (Lassiter & Goodman, 2017; Leffel et al., 2019; Xiang et al., 2022).

In this paper, we are interested in testing a specific prediction of the model we presented in Égré et al. (2023). Specifically, our model predicts a contrast between two kinds of *approximator* words (Sauerland & Stateva, 2007): the vague approximator “around” and the precise approximator “between” (as in “around 4” and “between 3 and 5”). Both types of expressions are used to convey uncertain numerical representations. But whereas “between” selects a fixed, explicit interval, “around n ” expresses that the value of interest lies in an interval centered on n but whose bounds are underspecified. A listener who interprets an “around” statement can reason that values closer to n are more likely to belong to the unknown interval than more remote values, leading to a posterior estimate of the variable of interest that favors values closer to n , while no such conclusion is warranted in the case of “between m and m' ”, apart from the exclusion of values outside the specified interval.

There is, in that regard, a similarity between this account of “around”, and the Bayesian account given by Tenenbaum and Griffiths (2001) of Shepard’s law of generalization (Shepard, 1987). According to Tenenbaum and Griffiths (2001), for some empirical term C whose extension is given to us by ostension through individual exemplars (such as “healthy”, “good to eat”), our inductive inference that some novel y is C after an observation that a given x is C is based on a representation of the various intervals that connect y and x within the relevant space, and it involves averaging out the weights put on those intervals as respective hypotheses. Similarly, our model in Égré et al. (2023) comes down to the idea that a vague expression like “around” causes a Bayesian interpreter to *average hypotheses* about the intervals denoted by “around n ” in a way that ends up putting more weight on values closer to n . This effect is not predicted for the precise approximator “between”, because its meaning is determinate.

However, while the generalization model is concerned with inferring the uncertain extension of a category based on some known exemplar, our model incorporates an additional level of lexical uncertainty: it targets inferences about an uncertain value (cardinality, age, or height) from lexical information which is also uncertain (the denotation of “around n ”, “tall”, or “old”). In the generalization model, the uncertain extension of a category consists in principle of *all* connected regions that include x along the relevant dimensions and are compatible with prior beliefs. In the case of a lexical expression such as “around” on the other hand, by assumption the uncertain interval is constrained by lexical meaning, namely to just the symmetric intervals centered on n . Similarly, for vague adjectives like “tall” or “old”, their possible denotations consist of only the semi-closed intervals above a given threshold (Kennedy, 2007; Lassiter & Goodman, 2017). Hence, while the denotation of these vague terms is uncertain, this uncertainty is lexically informed to ease and guide interpretation. We argued in Égré et al. (2023) that it is precisely thanks to their semantic underspecification that vague expressions manage to serve a specific communicative function: in certain situations, they allow speakers to communicate more fine-grained probabilistic information than any of their precise counterparts.

In what follows, we present a series of online posterior-elicitation experiments whose aim is to establish whether “around” is indeed more than a shorthand way of expressing a fixed implicit “between”, and to validate the insights that both kinds of expressions communicate uncertainty in different ways. In order to show this, Section 2 spells out the difference in meaning between the vague and precise approximators “around” and “between”, and presents the account we proposed in Égré et al. (2023) of the different updates they are predicted to occasion on a Bayesian listener. Section 3 presents the design of the experiments, and Section 4 the results. Section 5 presents some additional analyses of the participants’ data, and Section 6 draws broader lessons from the account.

2 Two kinds of approximators and their Bayesian effect

2.1 Around and Between

Approximators are modifiers such as “around”, “roughly”, “approximately”, “between”, “more than”, and “less than”. They constitute a subset of what Lakoff (1973) called “hedgers” (“words whose job it is to make things fuzzier or less fuzzy”). When applied to numerals, approximators end up denoting a wider range of possible values than the numeral they modify.

Thus, (1a) may be uttered if 17, 20, or 24 people came to the party. More specifically, “around n ” seems to denote an implicit interval which contains the numeral n itself, and which is further constrained by contextual and pragmatic considerations (Channell, 1994; Krifka, 2007; Lasnik, 1999; Solt, 2014). By contrast,

an expression like “between 15 and 25” in (1b) also denotes an interval, but it specifies explicit numerical bounds.

- (1) *How many people came to the party?*
 - a. **Around** 20 people came to the party.
 - b. **Between** 15 and 25 people came to the party.

We therefore assume the expression “between m and m' ”, where m and m' are numerals, to have the following meaning:

- (2) “ x is between m and m' ” is true iff $x \in [m, m']$.

For “around n ”, following Égré et al. (2023), we also assume its meaning to select an interval, but this interval is centered on n , and crucially the bounds of this interval are left open. A paraphrase for it would be: “ x is equal to n , give or take t ”, leaving t indeterminate:

- (3) “ x is around n ” is true iff $x \in [n - t, n + t]$

Under those assumptions, the Bayesian update occasioned by “between” is straightforward and described in (4). This update eliminates values outside the interval $[m, m']$, but crucially keeps the *relative* probabilities of the values within the interval as they were according to the prior (the notation $\propto$ indicates equality up to a multiplicative constant).

$$(4) \quad \mathbb{P}[x = k \mid x \text{ is between } m \text{ and } m'] \propto \begin{cases} \mathbb{P}[x = k] & \text{if } k \in [m, m'] \\ 0 & \text{if } k \notin [m, m'] \end{cases}$$

For “around”, on the other hand, the update rule produces a more complex effect (5): no value admitted by the prior is eliminated proper, but values closer to n will receive more weight than values further from n .¹ As illustrated in Figure 1, this is explained by the fact that the closer the value $x = k$ is to n , the more intervals of the form $[n - t, n + t]$ it belongs to. Put differently, under uncertainty about both x and t , we still know that the closer x is to n , the more likely it is to belong to the interval $[n - t, n + t]$. Hence, a Bayesian listener infers that x is more likely to be closer to n than further away when learning that x is around n .

$$(5) \quad \mathbb{P}[x = k \mid x \text{ is around } n] \propto \mathbb{P}[x = k] \sum_{i=|n-k|}^n \mathbb{P}[t = i]$$

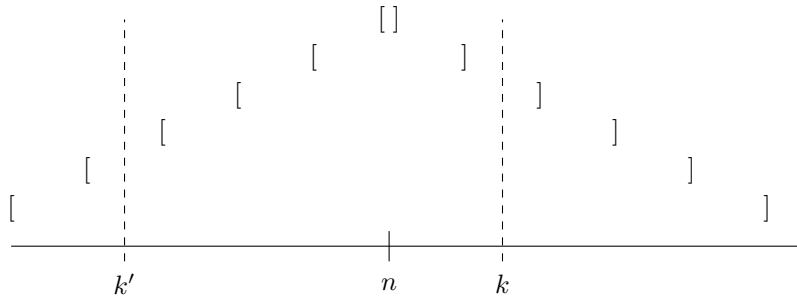


Figure 1: k (closer to n) belongs to four intervals centered around n , while k' (further from n) only belongs to two such intervals.

As noted above, the Bayesian effect of hearing “ x is around n ” on the estimate of x creates a gradient analogous to the one described by Tenenbaum and Griffiths (2001, Figure 1) in their account of ampliative inferences. Tenenbaum and Griffiths’s model describes how, upon learning that some observed value x is

¹See Égré et al. (2023), for a step-by-step derivation of this equation. Note that we use “Bayesian” effect in a broad sense throughout this paper: our listeners correspond to literal listeners in the sense of the RSA model of Goodman and Frank (2016), not to “Bayesian” listeners in the recursive sense elaborated in Égré et al. (2023).

in the category P , whose boundaries are uncertain, one can infer (with more or less confidence) that P also includes some new observed exemplar y . A single observation (such as “160 kg is what a lion can weigh”) is compatible with several intervals (the plausible intervals in which a lion’s weight lies), which all include the observation, and whose probabilities depend on the prior. The model for the interpretation of “around” too involves an inference under uncertainty about the boundaries of an unknown interval (“around n ”), based on the information that a certain value lies in it (the value n itself).

Yet the two models are quite different. In Tenenbaum and Griffiths’s model, the only uncertainty is about the true extension of an unknown interval C corresponding to a category (say the interval of possible weights for lions), whereas the values of x and y are observed. Furthermore, this uncertainty is about actual empirical regularities (e.g., the distribution of lions’ weights), and the prior probabilities of x and y are not independent from the prior probability over C . In contrast, in the model for “around”, both C , the denotation for the “around n ” expression, and x , the variable of interest, are unknown, and their prior distributions are independent. Importantly, in our case C is not about some concrete regularity in the world, it is an open parameter whose range is determined by the lexical meaning of the relevant expression, which thereby guides interpretation. As a result, while the models share a family resemblance, they are also mathematically distinct.²

2.2 Key prediction: the Ratio Inequality

A direct consequence of the different Bayesian updates produced by “around n ” and by “between m and m' ” (where n is in $[m, m']$) is the following. Consider two possible numerical values k and k' in the interval $[m, m']$ such that k is closer to n than k' is. Then the ratio of the posterior probabilities of k and k' is predicted to be greater for $\mathbb{P}[x = k \mid x \text{ is around } n]$ than for $\mathbb{P}[x = k \mid x \text{ is between } m \text{ and } m']$.

In other words, updating the prior with $u = \text{around } n$ leads to a greater ratio of posteriors, than updating the same prior with $u = \text{between } m \text{ and } m'$. This is formalized in (6), which Égré et al. call the Ratio Inequality. (6) does not depend on the prior distribution over k . It is therefore possible to test it on multiple listeners, who by default may have distinct priors.

$$(6) \quad \frac{\mathbb{P}[k \mid \text{around } n]}{\mathbb{P}[k' \mid \text{around } n]} \geq \frac{\mathbb{P}[k \mid \text{between } m \text{ and } m']}{\mathbb{P}[k' \mid \text{between } m \text{ and } m']} \quad (\text{Ratio Inequality})$$

In practice, there is a way to test (6)’s empirical validity in a more robust fashion, that is without having to arbitrarily choose which pair of numbers k and k' to use, and so as to filter out the noise of individual responses.³ There are multiple pairs (k, k') in $[m, m']$ such that k is closer to n than k' is (provided $i > 0$).⁴ It is therefore possible to average each side of inequality (6) over all possible such pairs, and to see whether on average the inequality is satisfied. A precise operationalization of this metric, and how we tested it, is what we present next.

3 Experimental paradigm

In this section, we spell out the general paradigm used in the three experiments presented in Section 4. This paradigm involves two main tasks for each expression: an INTERVAL task and a HISTOGRAM task. The dependent measure extracted from the latter task consists in a slight adaptation of the peakedness ratios introduced in (6). All three experiments will feature the same tasks and dependent variable. They will only differ in the arrangement of the critical items, and the presence or absence of fillers.

²In Tenenbaum and Griffiths’s model, the probability that a certain new exemplar y belongs to the uncertain interval C given that a previously observed value x is known to be in C is $\mathbb{P}[y \in C \mid x \in C] \propto \sum_{h: x, y \in h} \mathbb{P}[C = h]$ (weak sampling) or $\mathbb{P}[y \in C \mid x \in C] \propto \sum_{h: x, y \in h} \mathbb{P}[C = h] \frac{1}{|h|}$ (strong sampling). Our model assigns a probability to $X = k$, for any k , given that X is known to belong to the uncertain interval C . This yields $\mathbb{P}[X = k \mid X \in C] \propto \mathbb{P}[X = k] \sum_{h: k \in h} \mathbb{P}[C = h]$.

³A fixed ratio can be assumed to involve the same kind of Gaussian noise in its numerator and its denominator ($r = \frac{\mu + \epsilon}{\mu' + \epsilon'}$, with $\epsilon, \epsilon' \sim \mathcal{N}(0, \sigma)$). Assuming $\sigma \ll \mu, \mu'$, the overall noise will be approximately Gaussian too ($\simeq \frac{1}{\mu} \epsilon + \frac{\mu}{(\mu')^2} \epsilon'$). This kind of noise should be reduced by averaging.

⁴As a function of $i \geq 0$, this number of pairs $f(i)$ is equal to $2 \times \left(\binom{i+1}{2} + \binom{i}{2} \right) = 2i^2$.

3.1 Tasks

In order to collect empirical posterior distributions corresponding to $\mathbb{P}[k \mid \text{around } n]$ and $\mathbb{P}[k \mid \text{between } m \text{ and } m']$, and test (6)’s validity, we designed a web-based posterior-elicitation experiment. Participants were presented with a short written context, followed by a critical statement containing an expression of the form “around n ” or “between m and m' ”. Each participant was tested on one critical pair of expressions of the form “around n ” / “between m and m' ”. The main challenge, from an experimental perspective, was to ensure that the values n , m , and m' were such that the “around n ” and the “between m and m' ” expressions presented to each participant would give rise to elicited posterior distributions sharing the same support (where the support of a distribution is the set of values that are assigned a non-null probability). Even if the theory we proposed in Égré et al. (2023) does not predict that the two posteriors must share the same support, the comparison of the average ratio scores across the two distributions is *a priori* more meaningful if the two distributions have the same support, so that the number of pairs to be averaged is the same, and so that the differences in peakedness that we may find are not simply explained by the fact that one distribution is more spread out than the other.

Values of n were randomly set to 40, 50, or 60 for each participant. Values of m and m' were dynamically determined for each specific participant, in a way that will be made clear in the next paragraphs. This was meant to ensure each participant was exposed to “around” and “between” expressions whose induced posteriors shared the same support. The template used for the context and critical sentence is given in Figure 2.

Peter was at Mary’s party yesterday. He doesn’t know exactly how many people there were. He says:

There were {around n | between m and m' } people.

Figure 2: Template for the target sentence and its context.

A trial for a fixed “around n ” or “between m and m' ” expression was comprised of two main tasks: an INTERVAL task (see Figures 3a and 4a) and a HISTOGRAM task (see Figures 3b and 4b). For each expression, the INTERVAL task would always precede the HISTOGRAM task, and the INTERVAL task related to the “around” expression would always precede the one related to the “between” expression.

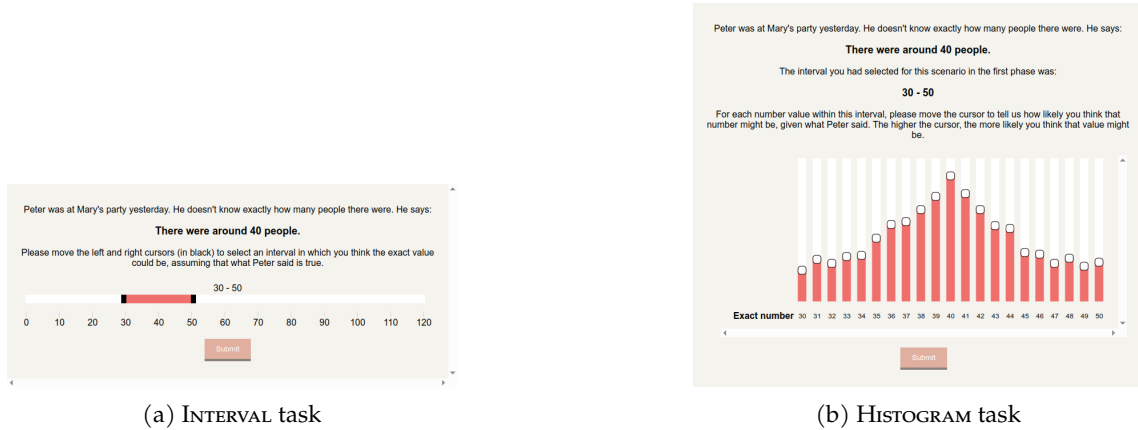
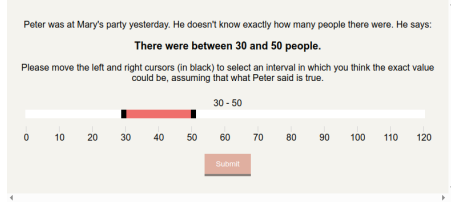
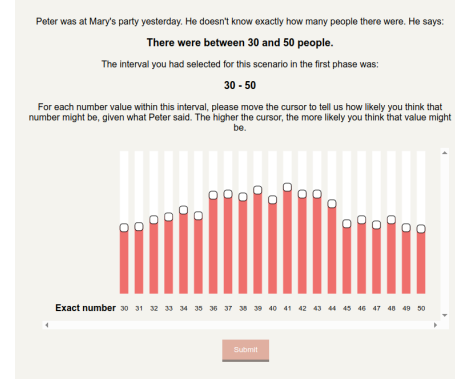


Figure 3: Screen captures of the two tasks for the target expression “around 40” (40 picked randomly from the set $\{40, 50, 60\}$). The values that appear here were defined by the authors and not a participant.



(a) INTERVAL task



(b) HISTOGRAM task

Figure 4: Screen captures of the two tasks for the target expression “between 30 and 50”, (30 and 50 determined based on the interval selected in Figure 3a). The values that appear here were defined by the authors and not a participant.

The goal of the INTERVAL task was twofold. First, it was used to elicit the lower and upper bounds of the support related to the posterior of a given expression (“around” or “between”), according to each participant. To set these bounds, participants were asked to move two sliders on a horizontal bar representing the $[0, 120]$ interval (see Figures 3a and 4a). The two bounds were supposed to define the interval within which the exact numerical value should be, given the target sentence and its context.

The second goal of this task was to determine the values of m and m' to use in the “between” expression corresponding to each “around” expression, in order to obtain posteriors defined on similar supports for both expressions. To achieve this, the values of the lower and upper bounds returned by the participant during the INTERVAL task testing “around” were reused as respectively the m and m' values of the critical “between m and m' ” expression. For instance, if “around 40” gave rise to a support with bounds $[27, 52]$ according to a certain participant, the corresponding “between” expression presented to them would be “between 27 and 52”. This was meant to guarantee that the support of the posterior distributions induced by “around n ” and “between m and m' ” were the same at the participant level.

The HISTOGRAM task was designed to elicit the posterior distributions $\mathbb{P}[k \mid \text{around } n]$ and $\mathbb{P}[k \mid \text{between } m \text{ and } m']$ defined on the domains returned by the participant during the INTERVAL tasks. Participants were again presented with the critical sentence and its context, and were reminded of the interval returned in the preceding INTERVAL task. They were then asked to “draw” a posterior distribution as a histogram, by adjusting one vertical slider per possible number within the interval (see Figures 3b and 4b). Sliders allowed null values. As a result, the interval over which a participant defined a posterior distribution did not necessarily coincide with the actual support of that posterior. The resulting histogram was intended to reflect the likelihood of each possible value within the original interval, conditional on the critical approximation statement.

3.2 Dependent variable and prediction tested

In the experiments described in the next section, we tested an aggregate version of the Ratio Inequality (6). First, we averaged the ratios for all pairs (k, k') such that k is closer to n than k' is. Secondly, the pairs (k, k') such that either k or k' is a “salient” value – e.g. “40” in “around 40” or “30”, and “50” in “between 30 and 50” – were excluded from the set of pairs over which individual posterior ratios were averaged.⁵ These values were excluded because their very salience might affect the behavior of participants in an experimental setting, leading them to raise their weights compared to others, thereby creating a confounding

⁵The preregistration for Experiment 1 uses a slightly different definition of a salient value, which additionally includes the bounds of the posterior evoked by “between m and m' ” – potentially different from m and m' . We did not use this extended notion in the analysis reported here, preferring the definition given in the main text and used in our preregistrations for Experiments 2 and 3. The original analysis yields qualitatively equivalent results and is available on [OSF](#).

effect.⁶ Thirdly, differences in the supports S_a and S_b of the posteriors induced by respectively “around” and “between” – which were meant to be minimized by the paradigm presented in the previous section – were accommodated by only considering pairs (k, k') in their intersection $(S_a \cap S_b)$.

This yields inequality (7), where K is the set of all pairs (k, k') such that (i) k is closer to n than k' is, (ii) k and k' differ from either n, m or m' , and (iii) both belong to $S_a \cap S_b$.

$$(7) \quad \underbrace{\frac{1}{|K|} \sum_{(k, k') \in K} \frac{\mathbb{P}[x = k \mid x \text{ is around } n]}{\mathbb{P}[x = k' \mid x \text{ is around } n]}}_{R_a} \geq \underbrace{\frac{1}{|K|} \sum_{(k, k') \in K} \frac{\mathbb{P}[x = k \mid x \text{ is between } m \text{ and } m']}{\mathbb{P}[x = k' \mid x \text{ is between } m \text{ and } m']}}_{R_b}$$

We call R_a (for “around”-ratio) and R_b (for “between”-ratio) the left-hand side and right-hand side of inequality (7), respectively. Inequality (7) is the one we propose to test in the series of experiments presented in the next section.

4 Experiments

4.1 Experiment 1

The first experiment was conducted on 207 participants (determined *via* a power analysis based on pilot data), and [preregistered on OSF](#). Participants were US-based Amazon Mechanical Turk Master Workers with a success rate of at least 98% on previous MTurk tasks. Participants were excluded based on preregistered demographic factors and quality requirements. First, participants who signaled that they were not native speakers of English were excluded. The remaining participants who were 1.645 standard deviations faster than the average completion time (i.e., above the 95th percentile of the distribution if the distribution is Gaussian) were also excluded. Finally, participants who provided inconsistent intervals on critical items were excluded. Inconsistency was defined as non-overlap with $[m, m']$ in the case of “between m and m' ”, and as non-overlap with the central value n in the case of “around n ”. These conditions were relatively permissive, which meant that “around”/“between” pairs not sharing the exact same support were included. For instance, participants could select an interval different from $[m, m']$ (but overlapping with it) when presented with “between m and m' ”; this selected interval would by construction differ from the interval selected for “around n ”. Participants could also choose to assign zero weights to values inside the selected intervals (for both “around n ” and “between m and m' ”), again leading to mismatching supports, even in cases where the intervals matched.

Each trial was comprised of an INTERVAL task immediately followed by the corresponding HISTOGRAM task. In addition to the two critical trials, this experiment involved six filler expressions featuring “around”, “between”, and “almost”, combined with numbers of varying granularities. The target “around” and “between” items were presented at two fixed positions in the experiment, as 4th and 8th trial respectively. The positions of the filler items were randomized over the set of remaining positions. The choice to associate filler expressions with different levels of granularity was intended to conceal the link between the critical “around n ” expression (position 4) and the critical “between m and m' ” expression (position 8). Table 1 summarizes the design of this experiment.

⁶Furthermore, “around 40” might trigger, through competition with the unmodified form “40”, the inference that 40 is unlikely to be the exact value, because if it were, a fully informed speaker would use “40”. Such an effect would go against our prediction and is not captured in our model. A fuller model would include additional pragmatic inferences of this sort that we do not address here.

Expression	Position	Category
“around n ”	4	Critical
“between m and m' ”	8	Critical
“between 80 and 90”	random($\{1, 2, 3, 5, 6, 7\}$)	Filler
“between 13 and 24”	random($\{1, 2, 3, 5, 6, 7\}$)	Filler
“around 70”	random($\{1, 2, 3, 5, 6, 7\}$)	Filler
“around 36”	random($\{1, 2, 3, 5, 6, 7\}$)	Filler
“almost 60”	random($\{1, 2, 3, 5, 6, 7\}$)	Filler
“almost 24”	random($\{1, 2, 3, 5, 6, 7\}$)	Filler

Table 1: Trials for Experiment 1

Because the INTERVAL and HISTOGRAM tasks related to a fixed expression were set to be adjacent within a trial, and because of the dependency between the m and m' values of the critical “between m and m' ” expression and the critical trial testing “around n ”, the order of these two trials could not be randomized.

A two-tailed Sign test for matched pairs⁷ with a significance threshold of .05 was used to test the hypothesis $R_a \neq R_b$ within-participant, on a sample size of 161 (after all exclusions⁸). It turned out significant in favor of $R_a > R_b$ ($p = 1.03\text{e-}12$), in line with inequality (7). 125 participants (77.6% of the sample) were associated with a positive ratio difference (in line with the hypothesis), while 34 participants (21.1% of the sample) were associated with a negative ratio. Two participants were linked to equal ratios.

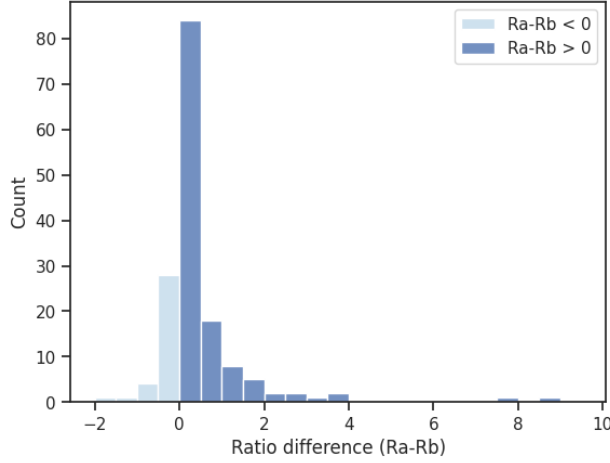
Exploratory two-tailed Wilcoxon tests performed on each subset of participants who were assigned to the same value of n turned out significant under Holm-Bonferroni correction ($p = 1.44\text{e-}4 < \frac{1\text{e-}3}{2}$ for $n = 40$; $p = 1.88\text{e-}6 < \frac{1\text{e-}5}{3}$ for $n = 50$; $p = 2.60\text{e-}4 < 1\text{e-}3$ for $n = 60$). Figure 5a illustrates the distribution of paired ratio differences.

Group-by-group averaged results are given in Figure 5b and confirm the within-participant results, in that posteriors induced by “around” tend to be more peaked than posteriors induced by a similar “between” expression.⁹

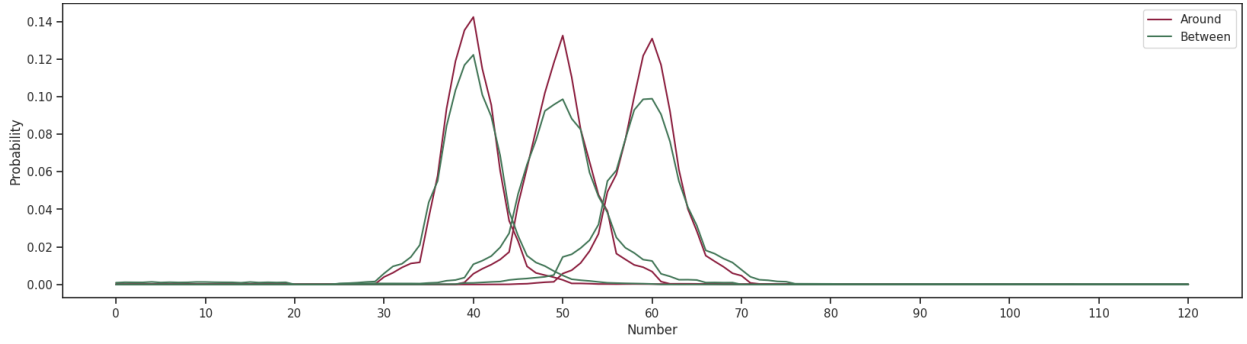
⁷The preregistration originally planned a two-tailed Wilcoxon signed-rank test, which turned out significant in favor of $R_a > R_b$ ($p = 6.04\text{e-}12$). We chose to report a Sign test instead (despite this test having less power), because we realized that a Wilcoxon test, being sensitive to the magnitude of individual ratio differences, may turn out significant globally by the effect of only one number group (e.g., $n = 60$) if this group is associated with high ratio differences in the expected direction.

⁸Breakdown: 7 participants excluded due to them being non-native speakers of English; 4 due to them being too fast; 9 due to a violation of the interval overlap condition; 26 because their target distributions did not allow the computation of an averaged peakedness score.

⁹When averaging posteriors across participants, we expect to get a peaked distribution even if each participant had reported a completely uniform distribution on the interval they had chosen. Imagine indeed that all participants report a uniform distribution, but choose intervals of different lengths. Since all intervals contain the target number (as a result of our exclusion criteria), numbers that are closer to the target will belong to more intervals than those that are further away. Therefore, when we average over participants, we expect that the closer a number is to the target, the more average weight it will get. For this reason, the ‘peakedness’ of the curves shown on Figure 5b cannot be interpreted as the average peakedness over participants, as it is likely to be more pronounced than in individual distributions. However, the visual contrast between “around” and “between” is still meaningful, since, given our design, each participant is expected to choose the same interval for “between” and “around”. The curves that result from restricting the data to the participants who chose exactly the same intervals for the two sentence types are visually similar to those we show here, and the same holds for comparable graphs in relation to subsequent experiments.



(a) Distribution of paired ratio differences ($R_a - R_b$).



(b) Averaged posteriors of the participants tested for $n = 40, 50$, or 60 .

Figure 5: Results of Experiment 1.

However, we also observed that participants tended to provide distributions that were more uniform as they advanced through the study.¹⁰ This constitutes a potential confound that we went on to address in two follow-up experiments (Experiment 2 and 3).

4.2 Experiment 2

Experiment 2 was [preregistered on OSF](#), and conducted on 150 participants.¹¹ Inclusion and exclusion criteria were the same as Experiment 1.¹²

In order to decrease the distance between the two critical trials and in turn reduce the “flattening” effect driven by the number of completed trials, fillers were removed. The design of this experiment is given in Table 2. Just like in Experiment 1, INTERVAL and HISTOGRAM tasks related to the same expression were set to be adjacent.

¹⁰Relevant code and plots are available on [OSF](#).

¹¹We adjusted the number based on the exclusion rate observed in Experiment 1.

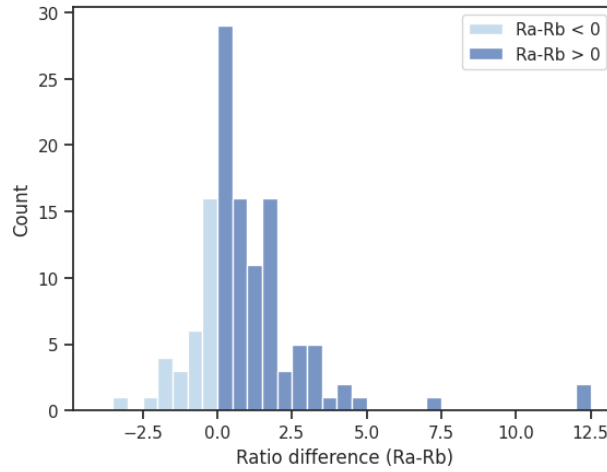
¹²These deviate slightly from the exclusion criteria preregistered for this experiment, but we preferred to adopt a uniform approach to exclusions across experiments. Using the preregistered criteria instead does not alter the reported results; this can be verified based on the code and data available on [OSF](#).

Expression	Position	Category
“around n ” (INTERVAL+HISTOGRAM)	1	Critical
“between m and m' ” (INTERVAL+HISTOGRAM)	2	Critical

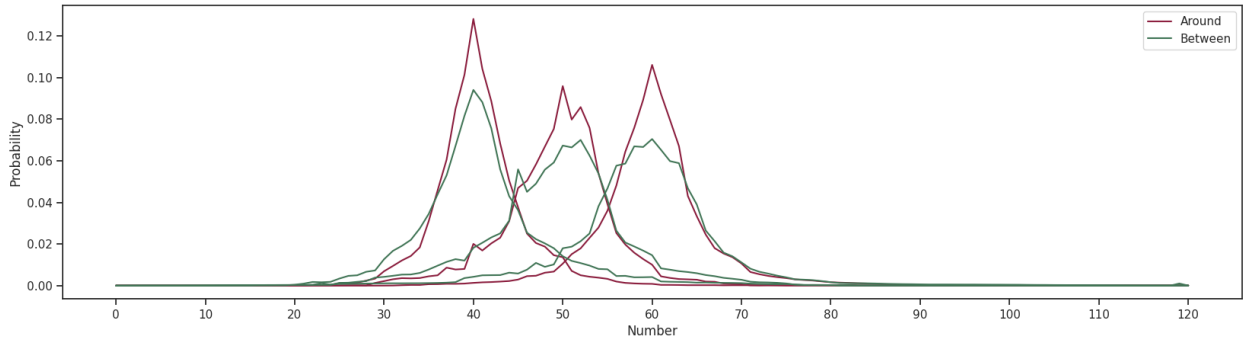
Table 2: Trials for Experiment 2

A preregistered one-tailed Sign test for matched pairs with a significance threshold of .05, applied to a sample of size 123 after all exclusions,¹³ turned out significant in favor of $R_a > R_b$ ($p = 1.66\text{e-}8$), again in line with inequality (7). 92 participants (74.8% of the sample) were associated with a positive ratio difference, while 31 participants (25.2% of the sample) were associated with a negative difference. Figure 6a illustrates the distribution of paired ratio differences.

Preregistered, number-specific Wilcoxon tests (one-tailed) ended up significant as well under Holm-Bonferroni correction ($p = 7.98\text{e-}4 < \frac{1\text{e-}2}{2}$ for $n = 40$; $p = 8.47\text{e-}4 < 1\text{e-}3$ for $n = 50$; $p = 7.78\text{e-}6 < \frac{1\text{e-}4}{3}$ for $n = 60$). By-number averaged results are displayed in Figure 6b and confirm the within-participant results.



(a) Distribution of paired ratio differences ($R_a - R_b$).



(b) Averaged posteriors of the participants tested for $n = 40, 50$, or 60 .

Figure 6: Results of Experiment 2.

Even though Experiment 2 corroborated the result of Experiment 1, its design did not allow us to totally remove the potential order effect between “around” and “between”, since the HISTOGRAM task related to

¹³Breakdown: 5 participants were excluded because their answers were not complete; 1 participant due to them being a non-native speaker of English; 2 due to them being too fast; 11 due to a violation of the interval overlap condition; 8 because their target distributions did not allow the computation of an averaged peakedness score.

“around” still systematically preceded the HISTOGRAM task related to “between”. This issue is what motivated Experiment 3.

4.3 Experiment 3

Experiment 3 was [preregistered on OSF](#) and conducted on 286 participants. The sample size was determined based on an a priori power analysis using as target effect size half the effect-size of Experiment 2 (which led to a target sample size of 250), and on the exclusion rates from Experiments 1 and 2. Inclusion and exclusion criteria were the same as for Experiments 1 and 2.¹²

Experiment 3 differs from Experiment 2 only in that the two HISTOGRAM tasks are fully randomized across participants. This was achieved by decoupling these tasks from their corresponding INTERVAL tasks.¹⁴ Specifically, Experiment 3 started with two (non-randomized) INTERVAL tasks, testing first “around n ”, and then “between m and m' ” (m and m' still being dynamically determined). The two INTERVAL tasks were then followed by the corresponding HISTOGRAM tasks, in a randomized order. Table 3 summarizes the design of this experiment.

Expression	Position	Category
“around n ” (INTERVAL)	1	Critical
“between m and m' ” (INTERVAL)	2	Critical
“around n ” (HISTOGRAM)	random($\{3, 4\}$)	Critical
“between m and m' ” (HISTOGRAM)	random($\{3, 4\}$)	Critical

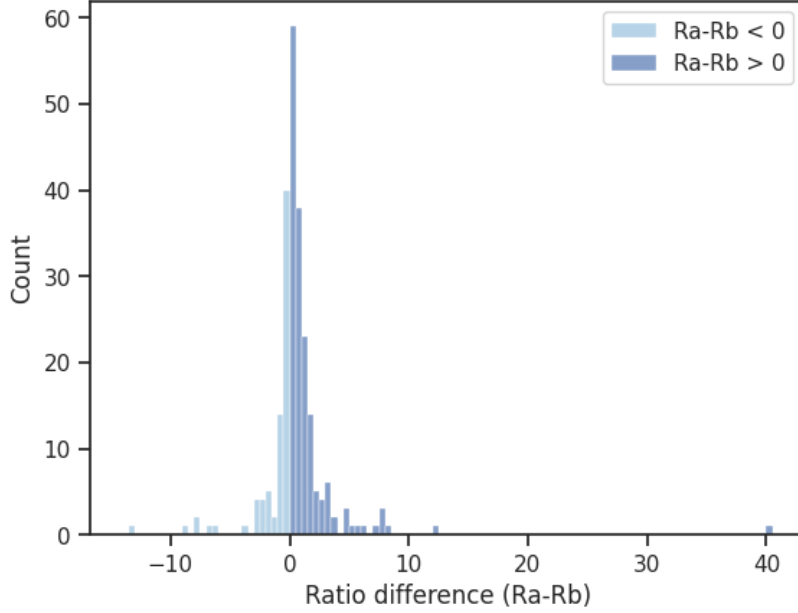
Table 3: Trials for Experiment 3

A preregistered one-tailed Sign test for matched pairs performed on a sample of 240 participants after all exclusions¹⁵ turned out significant in favor of $R_a > R_b$ ($p = 6.80\text{e-}9$). Figure 7a illustrates the distribution of paired ratio differences.¹⁶ Group-by-group averaged results are given in Figure 7b. In total, 164 participants (68.3% of the sample) were associated with a positive ratio difference, while 76 participants (31.7% of the sample) were associated with a difference going in the opposite direction.

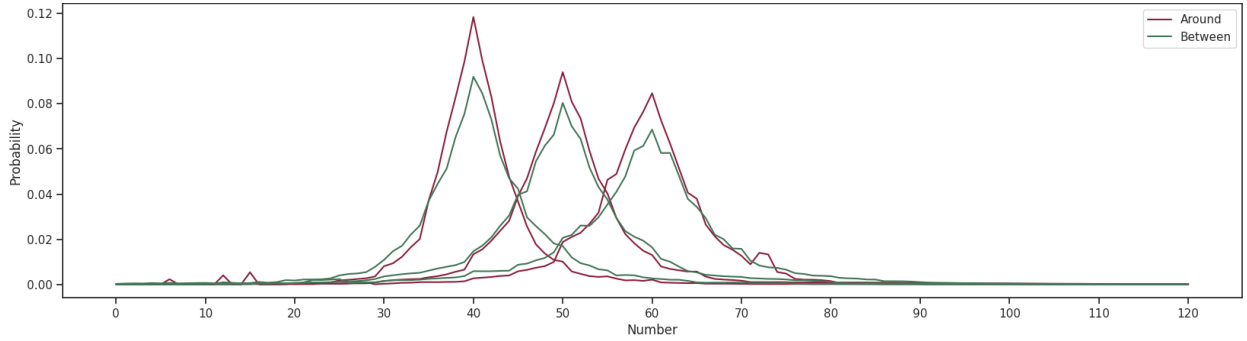
¹⁴We thank REDACTED for suggesting this modification.

¹⁵Breakdown: 3 participants were excluded due to them being non-native speakers of English; 2 due to them being too fast; 8 due to a violation of the interval overlap condition; 33 because their target distributions did not allow the computation of an averaged peakedness score.

¹⁶In this Experiment, we chose not to preregister additional number-specific one-tailed Wilcoxon tests. This is because power calculations suggested that the target effect size for such tests – which was expected to be reduced by the neutralization of order effects – could only be reached with significantly bigger sample sizes (> 450 , given past exclusion rates). For comparison with Experiments 1 and 2, we performed these tests in an exploratory way. They ended up significant under Holm-Bonferroni corrections ($p = 1.10\text{e-}5 < \frac{1\text{e-}4}{3}$ for $n = 40$; $p = 4.98\text{e-}4 < 1\text{e-}3$ for $n = 50$; $p = 3.23\text{e-}4 < \frac{1\text{e-}3}{2}$ for $n = 60$).



(a) Distribution of paired ratio differences ($R_a - R_b$).



(b) Averaged posteriors of the participants tested for $n = 40, 50$, or 60 .

Figure 7: Results of Experiment 3.

In sum, the fully randomized design of Experiment 3 confirms the results of both previous experiments while more carefully controlling for order effects.

5 Discussion

Our experiments used a within-participant design to test and confirm the peakedness contrast predicted by the Ratio Inequality (6) for pairs of “around” and “between” expressions. Our key result is that even when the bounds for “between” are set to coincide with those selected for “around”, there is a difference in the degree of belief attached to the values in those intervals. This difference supports the hypothesis that the interpretation of “around n ” involves probabilistic reasoning about the possible intervals it can denote, resulting in a posterior distribution over the variable of interest that is comparatively more peaked than the one obtained from a maximally similar “between m and m' ” expression.

In our study, we used the aggregate Ratio Inequality (7) to avoid having to rely on a single pair of numbers, so as to guard against noise in the data. Even with such an aggregate measure, we find that for between $1/5$ and $1/3$ of the participants, the difference in score between the two sentence types was of the

opposite sign as the one we predicted. This is not in fact particularly concerning. A sign-reversal rate of about 32% (Experiment 3) is not unexpected under the assumption that participants respond with some noise; assuming a Gaussian distribution of paired differences, such a proportion of sign reversals would correspond to a moderate effect size (specifically, 32% sign reversals are expected for an effect size of $d = 0.47$), and is therefore unexceptional.¹⁷

Another concern one might have is that even though “around” and “between” expressions were dynamically presented to each individual participant in order to be maximally similar in terms of the support of their induced posteriors, not all participants matched that behavior. To probe this, we conducted post hoc analyses (available on [OSF](#)) looking more closely at the properties of the *intervals* chosen by our participants during the INTERVAL task and of the *supports* of the corresponding posteriors reported in the HISTOGRAM task.

The proportion of participants with strictly matching intervals/supports for critical “around” and “between” trials varies between about a half to a third (Experiment 1: 53.4% with matching intervals; 50.9% with matching supports; Experiment 2: 35.8%/30.9%; Experiment 3: 33.8%/29.2%.) The remaining cases are split in two groups: a group in which the interval/support corresponding to “around” was smaller in size than the interval/support corresponding to the matching “between” expression, and a group in which the interval/support corresponding to “around” was larger in size than the interval/support corresponding to the matching “between” expression. The former group appears to be the most problematic *a priori*, because for this group, the “around”-posterior, being defined on a comparatively smaller interval, may mechanically be more peaked than the corresponding “between”-posterior, *independently* of our original prediction.¹⁸

Approximately 50% of the participants belonged to this potentially problematic group across experiments, with little variation depending on whether intervals or supports were considered. However, post-hoc linear mixed-effect regressions indicate that, across experiments, sentence type (“around” vs “between”) is a significant predictor of the peakedness ratio, even controlling for the potential effect of the length of the interval or support.¹⁹ This was done by fitting a LMER of the form $\log(R) \sim n + a + l + (1|i)$, with R the ratio variable, n the target number, a the approximator (“between” vs. “around”), l the length of the interval or the length of the support of the posterior, and i the participant.²⁰ Across experiments, comparisons between these models and simpler models without a as a factor always gave rise to a significant p -value, even after correction for multiple comparisons. This suggests that the effect we found is not specifically driven by the subset of the participants who associated “around” with a smaller interval/support than “between”. These results were maintained even when the dependency between log ratio and interval length was assumed to be quadratic or logarithmic instead of linear.²¹ This indicates that the effect of the approximator is robust across a variety of data analytic choices.

Lastly, restricting our initial analyses of the ratio differences to the uncontroversially unproblematic participants (i.e., those with matching intervals/supports for the two sentence types) consistently gives rise to significant p -values by a Sign test using Holm-Bonferroni correction for multiple comparisons.²² Overall, these analyses suggest that the slight variations observed in the intervals or supports of the relevant

¹⁷Let D be the random variable representing the difference of ratio-scores between “around” and “between”. If D follows a normal distribution with mean μ and standard deviation σ ,

$$\mathbb{P}[D > 0] = \mathbb{P}\left[\frac{D - \mu}{\sigma} > -\frac{\mu}{\sigma}\right] = \mathbb{P}\left[Z > -\frac{\mu}{\sigma}\right],$$

where $Z = \frac{D - \mu}{\sigma}$ and therefore follows a standard normal distribution. Since Z is symmetric around 0,

$$\mathbb{P}\left[Z > -\frac{\mu}{\sigma}\right] = \mathbb{P}\left[Z < \frac{\mu}{\sigma}\right] = \Phi\left(\frac{\mu}{\sigma}\right),$$

where Φ is the c.d.f of the standard normal distribution. Hence, $\mathbb{P}[D > 0] = 0.68$ iff $\Phi\left(\frac{\mu}{\sigma}\right) = 0.68$, i.e. if $d = \frac{\mu}{\sigma} = \Phi^{-1}(0.68) \approx 0.47$.

¹⁸For instance, if participants had a tendency, irrespective of the sentence type, to assign the maximal weight to the target, minimal weights to the bounds, with a linear decrease from the target to the bounds, shorter intervals would necessarily give rise to higher peakedness, corresponding to a greater ratio score.

¹⁹See the [OSF related material](#) for details of these analyses.

²⁰We therefore fitted 6 models in total, 2 for each experiment, depending on whether we used the support or the interval of the posterior as a predictor.

²¹Transformations suggested by an anonymous reviewer. These additional analyses are available on [OSF](#).

²²Experiment 1: 55/82 participants with matching supports were associated with ratio differences in line with the hypothesis; 59/86 if intervals were considered instead. Experiment 2: 29/38 (with supports) / 32/44 (with intervals). Experiment 3: 46/70 (with supports) / 54/81 (with intervals). Detailed implementation available on [OSF](#).

posteriors do not represent confounding factors when it comes to our assessment of the Ratio Inequality.

6 Conclusion

Language makes available different means to express uncertain magnitude estimates. We hypothesized, based on previous theoretical work (Égré et al. 2023), that vague expressions, and, in particular, vague approximators (as in “around n ”), are well-suited to convey approximate estimates which correspond to ‘peaked’ internal representations, while precise intervals (“between m and m' ”) tend to convey a range of magnitudes without specifically communicating peakedness. We provided experimental evidence supporting this hypothesis. Specifically, we have found evidence that even when the interval selected by “between” coincides for a listener with the interval selected by “around”, the probabilistic profiles of those intervals differ in the way we predicted.

This difference supports the hypothesis that the meaning of “around” induces a distinctive update profile, where uncertainty about the denoted interval produces a gradient effect, through a process of hypothesis averaging which is similar (though not identical) to the mechanisms involved in inferring the extension of a category from a small number of exemplars (cf. Tenenbaum and Griffiths’s 2001 theory of generalization).

Not only do these results support the theory we presented in Égré et al. (2023), they also provide a non-trivial confirmation of the Bayesian approach to linguistic interpretation of vague utterances more generally. There is in this regard a kind of parallelism between non-linguistic inferences under uncertainty about some variable (for instance based on the observation of an exemplar of a category, or on some approximate magnitude estimate) and linguistic inferences triggered by vague expressions, which makes them particularly useful to express imprecise magnitude estimates.

One limitation of this study regarding the phenomenon of interest is that we have only looked at how “around” and “between” are interpreted, not at how they are produced. The study does not tell us in which cases a speaker would prefer to use “between” or to use “around”. The theory we presented in Égré et al. (2023) considers this problem, and predicts that an agent might prefer to communicate their uncertain numerical estimate of a quantity using “around” when the speaker’s estimate is itself biased toward a specific value, and also when they want to convey some information about the underlying shape of their probability distribution. We leave to further work an experimental investigation of the production of distinct approximation expressions and their competition in different contexts.

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Contributions

PE, SV, and BS jointly defined the research project. AHM, SV, PE, and BS created the experimental designs together. AHM implemented the experimental designs and collected the data. Statistical analyses were planned by all four authors, and implemented by AHM. AHM, SV, PE, and BS wrote the paper together, PE and BS wrote the revised version of the paper, which all authors discussed. BS and PE coordinated the project. PE and BS share last authorship.

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